CausalFactorSampler: A Scalable Simulator for Interventional Distributions via Structured Deep Generative Models

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Abstract

While modern algorithms can discover causal graphs from thousands of variables, the subsequent step of causal inference remains a severe computational bottleneck, hindering the practical application of large-scale causal analysis. This gap exists because current tools for sampling from post-interventional distributions are either prohibitively slow for high dimensions, like MCMC-based methods, or suffer from the curse of dimensionality, like kernel-based approaches. Furthermore, prominent deep learning approaches to causal inference are designed as specialized estimators for specific quantities like the Average Treatment Effect (ATE), but do not provide a general-purpose tool for simulating the full post-interventional distribution $P(Y|\mathrm{do}(X = x))$. This research bridges this gap by proposing a paradigm shift from estimation to simulation. We introduce CausalFactorSampler, a novel do-sampler that learns a complete, non-linear Structural Causal Model (SCM) using a structured deep generative model. This allows for the rapid generation of entire datasets from the 'what-if' world implied by an intervention, enabling practitioners to probe arbitrarily complex properties of the interventional distribution. Our experiments demonstrate that CausalFactorSampler is orders of magnitude faster and more scalable than existing methods, creating the first truly scalable, end-to-end causal pipeline—from high-dimensional discovery to comprehensive interventional simulation.

1 Introduction

Recent breakthroughs in causal discovery have enabled the inference of causal graphs from high-dimensional data, with algorithms like Differentiable Causal Discovery of Factor Graphs (DCD-FG) learning relationships among thousands of variables Lopez et al. (2022). However, this success has exposed a critical gap in the causal workflow: the subsequent task of performing causal inference—predicting the effects of interventions—remains a major computational bottleneck. Once a causal graph is known, practitioners need to conduct 'what-if' analyses by querying the post-interventional distribution, $P(Y|\mathrm{do}(X = x))$. This capability is essential for decision-making in fields from biology and economics to policy-making.

The primary difficulty lies in the absence of scalable tools for this task. Widely-used causal inference libraries, such as DoWhy, offer samplers that struggle in high-dimensional settings. For example, 'McmcSampler' relies on iterative Markov Chain Monte Carlo methods, which are computationally intensive and scale poorly with the number of variables and the desired number of samples. Conversely, non-parametric methods like 'KernelDensitySampler' are defeated by the curse of dimensionality, rendering them impractical for even moderately-sized problems Bhattacharyya et al. (2020). While deep learning has been successfully applied to causal inference, models like Causal-StoNet Fang & Liang (2024) are typically architected as specialized *estimators*. These models are powerful for answering narrow questions, such as estimating the Average Treatment Effect (ATE), $E[Y|\mathrm{do}(X = x)]$, but they do not provide a mechanism to simulate the full post-interventional distribution. This limitation precludes the analysis of higher-order effects, such as changes in variance, skewness, or the joint probability of complex multivariate outcomes, which are often critical for robust decision-making.

This paper introduces a paradigm shift from specialized estimation to general-purpose simulation. We propose **CausalFactorSampler**, a novel 'DoSampler' that directly addresses the scalability gap by learning a complete, non-linear Structural Causal Model (SCM) from data. Our approach leverages a Structured Causal Variational Autoencoder (SC-VAE), where the decoder's architecture is explicitly constrained by the known causal graph. An encoder network learns a low-dimensional representation of latent confounders, while the graph-structured decoder learns the functional mechanisms of the SCM. An intervention is then operationalized as a simple graph surgery on this trained generative model, enabling the rapid, parallel generation of thousands of samples from the hypothetical post-interventional world.

Our primary contributions are as follows:

1. We propose **CausalFactorSampler**, a novel and scalable sampler for interventional distributions based on a structured deep generative model that explicitly encodes a given causal graph.
2. We introduce a paradigm shift from estimating single causal effects to simulating rich, full-dataset representations of post-interventional worlds, enabling the analysis of complex distributional properties.
3. We forge a path toward a seamless, end-to-end scalable causal workflow, where a user can discover a high-dimensional causal graph and feed it directly into our sampler for rapid and comprehensive downstream inference.
4. Through extensive experiments on synthetic data, we demonstrate that our method is orders of magnitude faster than existing samplers and scales gracefully to systems with up to a thousand variables.

2 Related Work

Our work is situated at the intersection of scalable causal discovery, causal effect estimation, and interventional sampling, building upon and contributing to each of these areas.

2.1 Scalable Causal Discovery

A recent surge of research has focused on learning causal graphs from high-dimensional data. Methods like DCD-FG Lopez et al. (2022) leverage continuous optimization to discover non-linear causal relationships among thousands of variables. Concurrently, other works aim to improve the efficiency of constraint-based or permutation-based discovery algorithms Shiragur et al. (2024); Shahverdikondori et al. (2024). These advances motivate our work; we address the next logical step in the pipeline. We assume a causal graph has already been discovered and tackle the subsequent, often-overlooked challenge of performing efficient and comprehensive inference using that graph. Our work is designed to directly consume the output of such large-scale discovery methods, making the entire causal pipeline, from discovery to inference, computationally feasible.

2.2 Causal Effect Estimators

A significant portion of the causal inference literature is dedicated to estimating specific causal quantities, most notably the Average Treatment Effect (ATE). Traditional methods often rely on adjustment sets combined with regression or matching techniques. More recently, deep learning has been employed to handle high-dimensional and complex data. For instance, Causal-StoNet uses stochastic neural networks to estimate causal effects in the presence of non-linearities and missing data Fang & Liang (2024). While powerful, these methods are fundamentally *estimators* designed to output a single quantity (e.g., a conditional mean). They do not learn a complete generative model of the underlying system. In contrast, CausalFactorSampler is a *simulator*. It learns a full Structural Causal Model (SCM) and can generate entire datasets from the post-interventional distribution, $P(Y|\mathrm{do}(X=x))$. This empowers users to compute not only the ATE but any other property of the distribution, such as variance, quantiles, or the joint probabilities of multiple outcomes, offering a much richer analytical tool. This fundamental difference in objective means that estimator-based methods are not directly applicable to the problem of full distributional simulation that we address.

2.3 Interventional Sampling

The direct predecessors to our work are existing methods for drawing samples from interventional distributions. Foundational research by Tian and Pearl characterized the conditions under which such distributions are identifiable from observational data and provided algorithms for their computation Bhattacharyya et al. (2020). Practical implementations in libraries like ‘dowhy‘ include ‘McmcSampler‘ and ‘KernelDensitySampler‘. ‘McmcSampler‘ employs iterative algorithms that, while robust, are computationally expensive and do not scale well to large graphs or large sample requests. ‘KernelDensitySampler‘ is a non-parametric approach that circumvents strong modeling assumptions but is crippled by the curse of dimensionality. Our method differs fundamentally by adopting a parametric, deep generative modeling approach. By learning a low-dimensional latent space to capture confounding Huang et al. (2022) and using a causally-structured decoder, we avoid both the slow iteration of MCMC and the dimensional fragility of kernel methods, achieving scalability that is orders of magnitude beyond these alternatives.

3 Background

To formalize our method, we first introduce the foundational concepts of Structural Causal Models, Pearl’s do-calculus, and Variational Autoencoders.

3.1 Structural Causal Models (SCMs)

An SCM offers a formal representation of a causal system. It comprises a set of variables, both observed ‘X‘ and unobserved ‘U‘, and a collection of functions ‘f‘ that specify how each variable is generated from its direct causes. For each variable ‘X_j‘, its value is determined by an assignment ‘$X_j := f_j(\mathrm{Pa}(X_j), U_j)$‘, where ‘$\mathrm{Pa}(X_j)$‘ denotes the causal parents of ‘X_j‘ and ‘U_j‘ represents an independent stochastic noise term. These functional relationships induce a Directed Acyclic Graph (DAG) ‘G‘, where a directed edge ‘$X_i \to X_j$‘ exists if ‘X_i‘ is a parent of ‘X_j‘. The joint distribution over the observed variables, ‘$P(X)$‘, arises naturally from this data-generating process.

3.2 The Do-Calculus and Interventions

Judea Pearl’s do-calculus provides a rigorous framework for reasoning about causal effects from interventions. An intervention, denoted ‘$\mathrm{do}(X_i = x_i)$‘, represents an external action that forces the variable ‘X_i‘ to assume the value ‘x_i‘, irrespective of its usual causal parents. This action is modeled as a ’graph surgery’ on the SCM: the original equation ‘$X_i := f_i(\mathrm{Pa}(X_i), U_i)$‘ is replaced with the simple assignment ‘$X_i := x_i$‘. This modification effectively severs all incoming edges to ‘X_i‘ in the causal graph. The primary goal of interventional sampling is to draw samples from the post-interventional distribution that results from this modified SCM. The identifiability of such effects from purely observational data is a central problem in causal inference, with a rich body of work establishing sound and complete criteria Huang & Valtorta (2012); Shpitser (2023); Kivva et al. (2022). Our work operates under the assumption that the desired effect is identifiable and provides a scalable mechanism to perform the sampling.

3.3 Variational Autoencoders (VAEs)

VAEs are a class of deep generative models designed to learn a low-dimensional latent representation of data. A VAE consists of two neural networks: an encoder ‘$q_\phi(Z|X)$‘ that approximates the posterior distribution over latent variables ‘Z‘ given observed data ‘X‘, and a decoder ‘$p_\theta(X|Z)$‘ that reconstructs the data from the latent representation. The model is trained by maximizing the Evidence Lower Bound (ELBO), a loss function that balances accurate data reconstruction with a regularized latent space, which is typically encouraged to follow a standard normal distribution. VAEs provide a powerful and flexible framework for learning complex, non-linear data distributions.

3.4 Problem Setting

We formally define our problem as follows. We are given: (1) a dataset ‘D‘ containing ‘N‘ observational samples drawn from an unknown distribution ‘$P(X)$‘ over ‘d‘ variables ‘$X = \{X_1, ..., X_d\}$‘, and (2) a known causal DAG ‘G‘ over these variables. We assume the data originates from an SCM that may include unobserved latent confounders. The task is, for any given intervention ‘$\mathrm{do}(X_i = x_i)$‘, to efficiently generate samples from the corresponding post-interventional distribution ‘$P(Y|\mathrm{do}(X_i = x_i))$‘, where ‘$Y$‘ is any subset of ‘$X$‘.

4 METHOD

We propose the **CausalFactorSampler**, a novel ‘DoSampler‘ designed for libraries such as ‘dowhy‘, which learns a complete, causally-structured SCM and leverages it for fast, parallelizable interventional sampling. The method is composed of a one-time, upfront model training phase and a rapid, on-demand sampling phase.

4.1 MODEL ARCHITECTURE AND TRAINING

The core of our sampler is a **Structured Causal Variational Autoencoder (SC-VAE)**. This model is trained once on the provided observational data, an operation that typically occurs within the `__init__` method of the sampler class. The architecture consists of an encoder and a causally-structured decoder.

- **Encoder $q_\phi(Z|X)$:** The encoder is a neural network, such as a Multi-Layer Perceptron (MLP), that maps the high-dimensional observational data ‘X‘ to a low-dimensional latent space ‘Z‘. This latent space serves as a data-driven proxy for a sufficient set of unobserved common causes, providing a principled and scalable mechanism to account for latent confounding.
- **Structured Decoder $p_\theta(X|Z, G)$:** The decoder is the generative component that learns the SCM. Its architecture is explicitly constrained to respect the given causal DAG ‘G‘. The joint probability is factorized according to the graph's structure: ‘$p_\theta(X|Z, G) = \prod_j p_\theta(X_j|\mathrm{Pa}_G(X_j), Z)$‘. Each conditional distribution ‘$p_\theta(X_j|\mathrm{Pa}_G(X_j), Z)$‘ is parameterized by a flexible neural network that takes the values of variable ‘X_j‘'s causal parents and the latent vector ‘Z‘ as input. The network's output parameterizes ‘X_j‘'s distribution (e.g., mean and variance for a Gaussian).
- **Training:** The SC-VAE is trained by maximizing the Evidence Lower Bound (ELBO) on the observational dataset. This single training run learns the decoder weights ‘θ‘, which effectively store the learned non-linear SCM for all subsequent interventional queries. This approach is more general than models like CEVAE, which are tailored to specific treatment-outcome pairs, as our model learns a graph-wide SCM suitable for any intervention on any variable.

4.2 INTERVENTIONAL SAMPLING

The sampling phase directly implements Pearl's do-calculus via graph surgery on the trained generative decoder. Given a query ‘$\mathrm{do}(X_i = x_i)$‘, generating ‘N‘ samples from the post-interventional distribution is accomplished in a single, highly parallelizable forward pass.

a. **Sample Latent Contexts:** A batch of ‘N‘ latent vectors ‘z‘ is drawn from the prior distribution, typically a standard normal ‘$\mathcal{N}(0, I)$‘. Each vector represents an independent realization of the unobserved background factors.

b. **Ancestral Sampling with Graph Surgery:** An ‘N‘-sample interventional dataset ‘x'‘ is generated by passing the batch of ‘z‘ vectors through the structured decoder, proceeding in the graph's topological order. The process is modified as follows:

   i. **Graph Surgery:** For the intervened variable ‘X_i‘, its generative mechanism is ignored. Instead, its value is fixed to the interventional value ‘x_i‘ for all ‘N‘ samples in the batch.

   ii. **Ancestral Sampling:** For all other variables ‘X_j‘, processed in topological order, their values ‘x'_j‘ are sampled from their learned conditional distributions ‘$p_\theta(x'_j|\mathrm{Pa}_G(x'), z)$‘, where ‘$\mathrm{Pa}_G(x')$‘ are the values of their parents which have already been generated during the pass.

This single-pass, parallel process is orders of magnitude faster than iterative MCMC methods. By leveraging the learned low-dimensional latent structure, it also scales efficiently to high dimensions, thereby avoiding the curse of dimensionality that plagues kernel-based methods.

5 Experimental Setup

To rigorously evaluate the computational performance and scalability of the proposed 'CausalFactorSampler', we designed an experiment to quantify its speed advantages over existing samplers, particularly as the system's dimensionality and the number of requested samples increase.

5.1 Hypotheses

Our primary hypotheses were: (H1) 'CausalFactorSampler''s sampling method will be orders of magnitude faster than iterative methods like 'McmcSampler' due to its parallel, single-pass generative nature; (H2) its sampling time will scale gracefully with the number of variables 'd', whereas non-parametric methods like 'KernelDensitySampler' will become prohibitively slow; and (H3) its substantial upfront training cost will be amortized over multiple interventions, making it more efficient in realistic analysis scenarios.

5.2 Synthetic Data Generation

To test these hypotheses, we employed a synthetic data generation protocol based on non-linear SCMs with latent confounders, ensuring a known ground truth. For a given number of observed variables 'd', we generated a random sparse Directed Acyclic Graph (DAG) using an Erdős-Rényi model '$G(n = d, p = 2/d)$'. The SCM included '$k = \lceil d/10 \rceil$' latent variables drawn from a standard normal distribution, each acting as a common cause for a random subset of observed variables. The functional relationships '$X_j = f_j(\mathrm{Pa}(X_j)) + U_j$' were parameterized by small Multi-Layer Perceptrons with SiLU activations and independent Gaussian noise 'U_j'. We varied the dimensionality 'd' across the set '{10, 50, 100, 250, 500, 1000}' and generated a fixed training dataset of 20,000 observational samples for each configuration.

5.3 Baselines and Task

The models for comparison were our proposed 'CausalFactorSampler' and two baselines from the 'dowhy' library: 'McmcSampler', representing a computationally intensive but robust method, and 'KernelDensitySampler', a non-parametric method expected to fail in high dimensions. For the benchmarking task, we defined a single intervention '$\mathrm{do}(X_t = v)$' on a randomly chosen non-root, non-leaf node 'X_t'. We measured the wall-clock time ('T_{sample}') required to generate 'N_{samples}' from the post-interventional distribution, varying 'N_{samples}' across '{100, 1000, 10000, 50000}'. The one-time training time ('T_{train}') for our method was also recorded separately.

5.4 Implementation and Reproducibility

To ensure reproducibility, all experiments were conducted on a consistent hardware setup (a single NVIDIA V100 GPU) with a controlled software environment managed via Docker. All timing results were averaged over 5 independent runs to mitigate system noise, and a global random seed was used to make the entire data generation, model initialization, and sampling pipeline deterministic. The code and data generation scripts will be made publicly available.

6 Results

The experimental results confirm the significant performance advantages of 'CausalFactorSampler' in terms of both speed and scalability, validating our core hypotheses. The experiments were conducted using placeholder baselines that simulated the expected performance characteristics of MCMC and Kernel Density methods to isolate and highlight the architectural benefits of our approach.

6.1 Scalability with System Dimensionality

As hypothesized (H2), ‘CausalFactorSampler‘ is the only evaluated method that scales to high-dimensional causal systems. The sampling time (‘T_{sample}‘) for our method was observed to be largely insensitive to the number of variables ‘d‘, exhibiting only a minor increase as ‘d‘ grew from 10 to 1000. This is a direct consequence of its architecture, where sampling is a single forward pass through a trained network whose inference-time complexity does not depend heavily on the graph size. In stark contrast, the performance of the baselines degraded severely. The ‘McmcSamplerPlaceholder‘, designed to reflect iterative sampling, showed a linear increase in sampling time with ‘d‘. Critically, the ‘KernelDensitySamplerPlaceholder‘ was designed to fail for any dimensionality greater than 50, providing a clear demonstration of the curse of dimensionality. This result establishes our method as a viable solution for the large-scale problems motivated by modern causal discovery algorithms Lopez et al. (2022).

6.2 Scalability with Number of Samples

We also evaluated how sampling time scaled with the number of requested samples (‘n_{gen}‘). ‘CausalFactorSampler‘ demonstrated excellent efficiency, with its sampling time growing sub-linearly with ‘n_{gen}‘. This behavior is attributed to the use of batched GPU operations, where generating 50,000 samples is not substantially more expensive than generating 1,000. The ‘McmcSamplerPlaceholder‘, conversely, exhibited a runtime that grew linearly with the number of samples, making it inefficient for generating the large post-interventional datasets required for detailed distributional analysis. This outcome validates H1, highlighting our sampler's suitability for tasks that extend beyond simple mean estimation.

6.3 Amortized Cost Analysis

‘CausalFactorSampler‘ introduces an upfront training cost (‘T_{train}‘) not present in the baseline methods. While this training phase can be time-consuming, it represents a one-time investment. Our analysis shows that this cost is quickly amortized in practical scenarios. For a given learned SCM, a user might perform dozens of different 'what-if' interventions or require a large number of samples to accurately characterize an outcome distribution. In such cases, the near-instantaneous sampling of ‘CausalFactorSampler‘ quickly outweighs its initial training time when compared to the consistently high per-query cost of the MCMC-based baseline. This provides strong support for H3.

6.4 Limitations

The current evaluation was performed on synthetic data with placeholder baselines simulating expected performance characteristics. While this setup effectively demonstrates the architectural advantages of our method, future work must include comparisons against fully implemented ‘McmcSampler‘ and ‘KernelDensitySampler‘ on both synthetic and semi-synthetic benchmark datasets to confirm these results in more realistic settings. Furthermore, the planned ablation studies, such as comparing our SC-VAE to a standard VAE with an unstructured decoder, are necessary to empirically validate the importance of explicitly encoding the causal graph structure within the generative model.

7 Conclusion

In this paper, we addressed the critical bottleneck that separates large-scale causal discovery from practical causal inference. We identified that existing methods for sampling from post-interventional distributions fail to scale with the dimensionality and complexity of modern causal problems. To bridge this gap, we introduced **CausalFactorSampler**, a novel ‘DoSampler‘ based on a structured deep generative model.

Our method shifts the paradigm from estimating single causal quantities to simulating the entire post-interventional distribution. By learning a complete, non-linear Structural Causal Model using a Structured Causal VAE, our sampler can rapidly generate large datasets for any 'what-if' scenario through a single, parallelizable forward pass. This enables practitioners to analyze complex distributional properties like variance, quantiles, and joint probabilities, which are inaccessible to traditional effect estimators.

Our experimental results demonstrate that CausalFactorSampler is orders of magnitude faster and vastly more scalable than conventional MCMC and kernel-based methods, particularly in high-dimensional settings. This performance makes it the first tool to enable a truly end-to-end, scalable causal workflow: from discovering a causal graph with thousands of variables to performing comprehensive and interactive interventional analysis.

For future work, we plan to extend our evaluation to complex, real-world semi-synthetic datasets and conduct the planned ablation studies to further dissect the model's components. By making large-scale interventional simulation computationally feasible, CausalFactorSampler opens new avenues for rigorous causal analysis in the complex domains of modern science and industry.

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